PORTFOLIO PERSPECTIVE: DIVERSIFICATION AND RISK REDUCTION

Learning Outcome

- describe the portfolio approach to investing

One of the biggest challenges faced by individuals and institutions is to decide how to invest for future needs. For individuals, the goal might be to fund retirement needs. For such institutions as insurance companies, the goal is to fund future liabilities in the form of insurance claims, whereas endowments seek to provide income to meet the ongoing needs of such institutions as universities. Regardless of the ultimate goal, all face the same set of challenges that extend beyond just the choice of what asset classes to invest in. They ultimately center on formulating basic principles that determine how to think about investing. One important question is: Should we invest in individual securities, evaluating each in isolation, or should we take a portfolio approach? By "portfolio approach," we mean evaluating individual securities in relation to their contribution to the investment characteristics of the whole portfolio. In the following section, we illustrate a number of reasons why a diversified portfolio perspective is important.

Historical Example of Portfolio Diversification: Avoiding Disaster

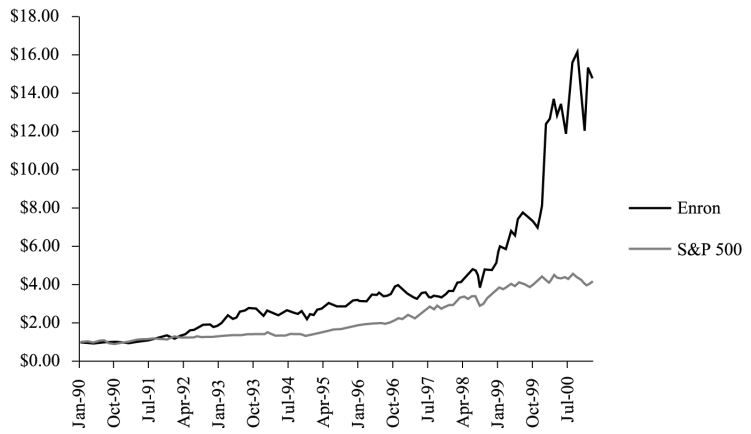
Portfolio diversification helps investors avoid disastrous investment outcomes. This benefit is most convincingly illustrated by examining what may happen when individuals have *not* diversified.

We are usually not able to observe how individuals manage their personal investments. However, in the case of US 401(k) individual retirement portfolios,[2] it is possible to see the results of individuals' investment decisions. When we examine their retirement portfolios, we find that some individual participants make sub-optimal investment decisions.

During the 1990s, Enron Corporation was one of the most admired corporations in the United States. A position in Enron shares returned over 27 percent per year from 1990 to September 2000, compared to 13 percent for the S&P 500 Index for the same time period.

Exhibit 1:

Value of US$1 Invested from January 1990 to September 2000 Enron vs. S&P 500 Composite Index (01/01/1990 = US$1.00)



returned over 27 percent per year from 1990 to September 2000, compared to 13 percent for the S&P 500 Index for the same time period.

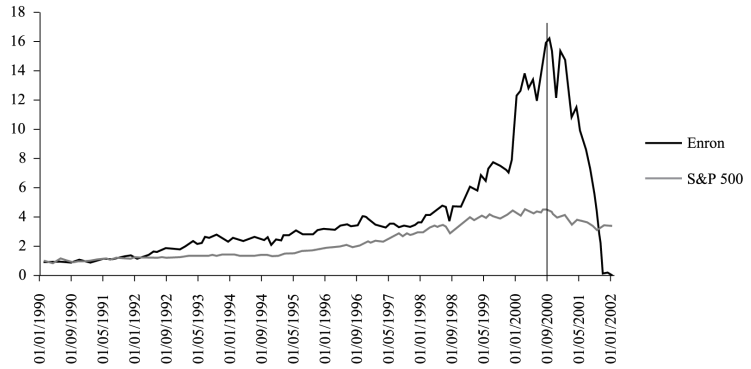
Source: Thomson Reuters Datastream.

During this time period, thousands of Enron employees participated in the company's 401(k) retirement plan. The plan allowed employees to set aside some of their earnings in a tax-deferred account. Enron participated by matching the employees' contributions. Enron made the match by depositing required amounts in the form of Enron shares. Enron restricted the sale of its contributed shares until an employee turned 50 years old. In January 2001, the employees' 401(k) retirement accounts were valued at over US$2 billion, of which US$1.3 billion (or 62 percent) was in Enron shares. Although Enron restricted the sale of shares it contributed, less than US$150 million of the total of US$1.3 billion in shares had this restriction. The implication was that Enron employees continued to hold large amounts of Enron shares even though they were free to sell them and invest the proceeds in other assets.

A typical individual was Roger Bruce,[3] a 67-year-old Enron retiree who held all of his US$2 million in retirement funds in Enron shares. Between January 2001 and January 2002, Enron's share price fell from about US$90 per share to zero.

Exhibit 2:

Value of US$1 Invested from January 1990 to January 2002 Enron vs. S&P 500 Composite Index (1/1/1990 = US$1.00)



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This graph shows the value of US$1 Invested from January 1990 to January 2002 for Enron vs. S&P 500 Composite Index (1/1/1990 = US$1.00). Value is in the Y-axis and month/year is in the X-axis. Enron's share price generally increased from January 1990 to December 2000 but between January 2001 and January 2002, Enron's share price fell from about US$90 per share to zero.

Source: Thomson Reuters Datastream.

Employees and retirees who had invested all or most of their retirement savings in Enron shares, just like Mr. Bruce, experienced financial ruin. The hard lesson that the Enron employees learned from this experience was to "not put all your eggs in one basket."[4] Unfortunately, the typical Enron employee did have most of his or her eggs in one basket. Most employees' wages and financial assets were dependent on Enron's continued viability; hence, any financial distress on Enron would have a material impact on an employee's financial health. The bankruptcy of Enron resulted in the closing of its operations, the dismissal of thousands of employees, and its shares becoming worthless. Hence, the failure of Enron was disastrous to the typical Enron employee.

Enron employees were not the only ones to be victims of over-investment in a single company's shares. In the defined contribution retirement plans at Owens Corning, Northern Telecom, Corning, and ADC Telecommunications, employees all held more than

The good news in this story is that the employees participating in employer-matched 401(k) plans since 2001 have significantly reduced their holdings of their employers' shares.

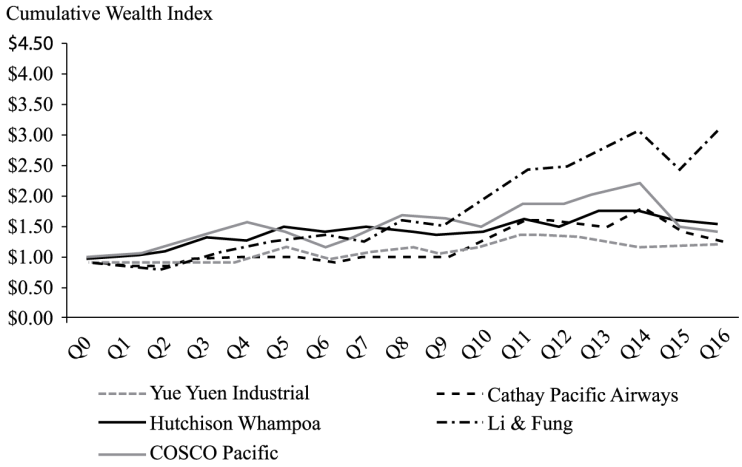
Thus, by taking a diversified portfolio approach, investors can spread away some of the risk. Rational investors are concerned about the risk–return trade-off of their investments. The portfolio approach provides investors with a way to reduce the risk associated with their wealth without necessarily decreasing their expected rate of return.

Portfolios: Reduce Risk

In addition to avoiding a potential disaster associated with over investing in a single security, portfolios also generally offer equivalent expected returns with lower overall **volatility** of returns—as represented by a measure such as standard deviation. Consider this simple example: Suppose you wish to make an investment in companies listed on the Hong Kong Stock Exchange (HKSE) and you start with a sample of five companies.[5] The cumulative returns for 16 fiscal quarters are shown in Exhibit 3.

Exhibit 3:

Cumulative Wealth Index of Sample of Shares Listed on HKSE (initial amount= US$1.00)



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This graph shows the Cumulative Wealth Index for 16 fiscal quarters of a sample of five companies listed on the Hong Kong Stock Exchange (initial amount = US$1.00). The five companies are Yue Yuen Industrial, Hutchison Whampoa, COSCO Pacific, Cathay Pacific Airways, and Li & Fung. Cumulative Wealth Index is in the Y-axis and fiscal quarter is in the X-axis.

Source: Thomson Reuters Datastream.

The individual quarterly returns for each of the five shares are shown in Exhibit 4. The annualized means and annualized standard deviations for each are also shown.[6]

Exhibit 4:

Quarterly Returns (in Percent) for Sample of HKSE Listed Shares over 16 Fiscal Quarters

	Yue Yuen Industrial	Cathay Pacific Airways	Hutchison Whampoa	Li & Fung	COSCO Pacific	Equally Weighted Portfolio
Q1	–11.1%	–2.3%	0.6%	–13.2%	–1.1%	–5.4%
Q2	–0.5	–5.4	10.8	1.7	21.0	5.5

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	Industrial	Airways	Whampoa	Fung	Pacific	Portfolio
Q3	5.7	6.8	19.1	13.8	15.5	12.2
Q4	5.3	4.6	–2.1	16.9	12.4	7.4
Q5	17.2	2.4	12.6	14.5	–7.9	7.8
Q6	–17.6	–10.4	–0.9	4.4	–16.7	–8.2
Q7	12.6	7.4	4.2	–10.9	15.4	5.7
Q8	7.5	–0.4	–3.6	29.2	21.9	10.9
Q9	–7.9	1.3	–5.1	–2.0	–1.6	–3.1
Q10	8.2	27.5	0.1	26.0	–10.1	10.3
Q11	18.3	24.3	16.5	22.8	25.7	21.5
Q12	0.1	–2.6	–6.7	–0.4	0.3	–1.8
Q13	–6.2	–4.2	16.7	11.9	11.1	5.8
Q14	–8.0	17.9	–1.8	12.4	8.4	5.8
Q15	3.5	–20.1	–8.5	–20.3	–31.5	–15.4
Q16	2.1	–11.8	–2.6	24.2	–6.1	1.2
Mean annual return	7.3%	8.7%	12.3%	32.8%	14.2%	15.1%
Annual standard deviation	20.2%	25.4%	18.1%	29.5%	31.3%	17.9%
Diversification ratio						71.9%

Source: Thomson Reuters Datastream.

Suppose you want to invest in one of these five securities next year. There is a wide variety of risk–return trade-offs for the five shares selected. If you believe that the future will replicate the past, then choosing Li & Fung would be a good choice. For the prior four years, Li & Fung provided the best trade-off between return and risk. In other words, it provided the most return per unit of risk. However, if there is no reason to believe that the future will replicate the past, it is more likely that the risk and return on the one security selected will be more like selecting one randomly. When we randomly selected one security each quarter, we found an average annualized return of 15.1 percent and an average annualized standard deviation of 24.9 percent, which would now become your expected return and standard deviation, respectively.

Alternatively, you could invest in an equally weighted portfolio of the five shares, which means that you would invest the same dollar amount in each security for each quarter. The quarterly returns on the equally weighted portfolio are just the average of the returns of the individual shares. As reported in Exhibit 4, the equally weighted portfolio has an average return of 15.1 percent and a standard deviation of 17.9 percent. As expected, the equally weighted portfolio's return is the same as the return on the randomly selected security. However, the same does not hold true for the portfolio standard deviation. That is, the standard deviation of an equally weighted portfolio is not simply the average of the standard deviations of the individual shares. In a more advanced reading we will demonstrate in greater mathematical detail how such a portfolio offers a lower standard deviation of return than the average of its individual components due to the correlations or interactions between the individual securities.

Because the mean return is the same, a simple measure of the value of diversification is calculated as the ratio of the standard deviation of the equally weighted portfolio to the standard deviation of the randomly selected security. This ratio may be referred to as the **diversification ratio**. In this case, the equally weighted portfolio's standard deviation is approximately 72 percent of the average standard deviation of the 5 stocks (24.9%). The diversification ratio of the portfolio's standard deviation to the individual asset's standard deviation measures the risk reduction benefits of a simple portfolio construction method, equal weighting. Even though the companies were chosen from a similar industry grouping, we see significant risk reduction. An even greater portfolio effect (i.e., lower diversification ratio) could have been realized if we had chosen companies from completely different industries.

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associated with investing in a single company's shares. In this section we extended the notion of risk reduction through portfolios to illustrate why individuals and institutions should hold portfolios.

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